Medical Device Change Control Implementation

Nicolás Figueroa – Senior Validation Administrator
Hospira Costa Rica

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- > 2600 Employees
- Operated 24/7
- 3 Products:
  - Gravity Sets
  - Injectable Syringes
  - Infusion Pumps
- > 900 Validated Items (Equipment/Processes/Systems, etc)

"Faced with the choice between changing one's mind and proving that there is no need to do so, almost everyone gets busy on the proof."
— John Kenneth Galbraith

"Change is hard because people overestimate the value of what they have—and underestimate the value of what they may gain by giving that up."
— James Belasco and Ralph Stayer
Flight of the Buffalo (1994)

"After living with their dysfunctional behavior for so many years (a sunk cost if ever there was one), people become invested in defending their dysfunctions rather than changing them."
— Marshall Goldsmith
Mojo

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Medical Device Change Control

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General Requirements for Change Control of Medical Devices

- 21 CFR Part 820
  - 820.30 Design Controls
- Software Validation Guideline
  - Whenever software is changed, a validation analysis should be conducted not just for validation of the individual change, but also to determine the extent and impact of that change on the entire software system.
- ISO13485
  - Control of design and development changes

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Medical Device Manufacturing Elements Subject to Change Control

- Equipments
- Facilities and Utilities
- Processes
- Documentation
- Computer Systems / Software
- Product Design Changes

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Define the problem

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Problems that can occur with a poor change control process

- Inadequate change control exposes a company to product liability actions, results in product recalls, causes internal confusion, and is a serious violation of the Quality System (QS) regulation. - FDA

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Problems that can occur with a poor change control process

- Full extent of impacted elements in a change are not identified or considered.
- Impact is not adequately evaluated.
- Deliverables are not clearly identified.
- Lifecycle is not controlled (validations).
- Not all stakeholders or SMEs are involved in the process (Review Board).

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Establish the team to work on the new process

- Map your process to know which departments / functions / areas are involved.
- Working with lots of people can be difficult, manage a core team (key SMEs) and then extended teams for additional input/feedback on a prototype.
- Pass down impacted functional areas to the sites so they can organize workgroups as well for analysis and implementation of the changes.

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Identify the requirements for a proper change control process

- Regulatory Sources:
  - 21 CFR Part 820
  - Process Validation Guideline
  - GAMP
  - ICH Q9
- Procedures:
  - Current procedures need to be assessed as part of the requirements since they identify specific items to consider for each site or function that may need to be followed there.
- Business Requirements:
  - Metrics

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Performing a gap assessment against the current process

- Design a matrix that will map your current process against the requirements identified in the previous stage.
- The matrix should be a living document during the design of the new process.
- Once the new process is designed, pass down to impacted sites as follow up matrix to make sure all items are covered in local procedures.
- Review initial gap assessment with core team to ensure all impacted elements are covered.

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Create the new process

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Design of the new process

- Periodic meetings with the core team.
- Make sure to analyze each requirement thoroughly before trying to identify the current gap with it.
- Take into account the possible implications of changing the corporate procedures / guidelines in each site.
- Decide if it is going to be a centralized system or if there should be several sub-systems.
- Map a software into your process if you´re using one.
- Establish possible metrics of the process.

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Elements to modify for the new process

- Corporate Policies
- Corporate Procedures / Guidelines
- Local Policies
- Local Procedures / Guidelines
- Templates
- How to documents

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Identify Stakeholders and Key Players for implementation

- During implementation you want to focus more on identifying the SMEs that will lead each particular site team.
- Also, have someone at the sites look for all people that is impacted by the change control process so they can get involved in the implementation team.
- Communication is key !!! If you don´t get the buying from key players, you will end up with a lot of unexpected delays in the implementation process.

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Provide training to sites and functions

- Training is essential for a successful site implementation.
- First step is to communicate and explain the changes performed in the corporate policies and procedures to the SMEs.
- Second, train local experts on the mechanisms to perform the local procedures gap assessment.
- Third, point them on what key elements should they be focusing on and provide some ideas on how to implement that locally.

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Implement the new process

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Conduct site and function gap assessments

- Once the new process is established, the next step is to take the original gap assessment performed for the corporate policies and procedures and use that as the basis for the site gap assessments.
- First thing to determine is which procedures are impacted by the change.
- Also, determine if there should be mapping of the local processes.
- Process mapping is necessary to conduct process step gap assessment, the local process should have modules that map directly to the corporate process.

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Revise site and function procedures to align with the new process

- All impacted site procedures should get aligned with the new process.
- Also, conduct a broader search to identify possible indirectly – impacted procedures that may need modification as well.
- One of the most important things is try to maintain the “status quo” while deploying the changes: if the local or current process / procedure already has a similar process, consider keeping it or adapting it to change as little as possible.

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Train and roll out site and function procedures

- When the new process has been mapped locally in the impacted sites, train end users in the new process.
  - Keep track of feedback as there may be specific items that may not have been considered up to this point, there is still time to ammend.
- Perform a soft launch or demo with involved teams to identify possible clarifications required for the new process.

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Follow up on new process

- Lessons learned from implementation process
  - Document them!
- Perform Status Update Meetings with both the core and extended teams.
- Capture and interpret the metrics generated by the new system to identify improvement points.
- Survey users and impacted sites to evaluate the reaction towards the new process.

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Interactive Exercise - Conducting the Gap Analysis

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What to be looking for in your current process

- Do you want your process to be centralized or function – independent?
- Do you want it to be fully automated? Can you define URS for it?
- Who are the key players and SMEs involved in your process?
- Which are the improvement portions of your current process?
- Identify the requirements for your products / processes. Do you need to cover some regulatory gaps?

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Thank You

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